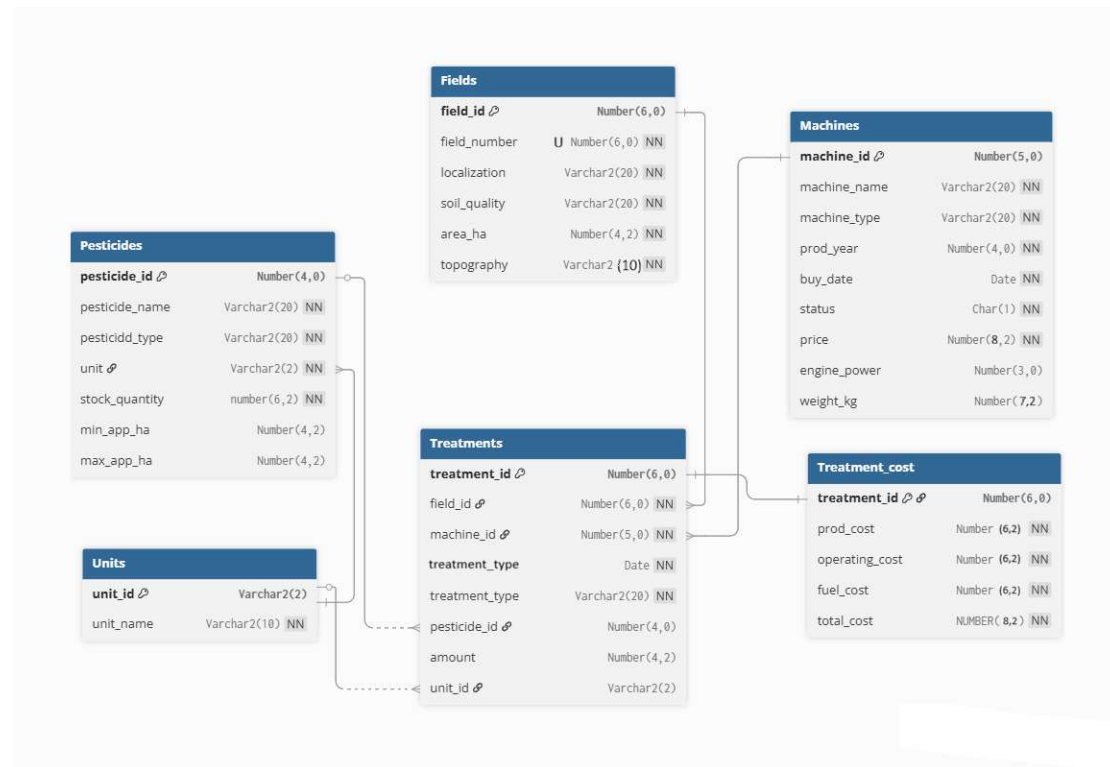


Treatment Records Database

This project presents the structure and functionality of a relational database designed for a modern agricultural enterprise. The primary goal of the system is to provide comprehensive management of agrotechnical processes, monitor hardware resources, and perform precise analysis of operational costs related to farming.

Relational logical model



Most frequent queries:

1. Queries regarding, for example, finances and cost management. For example, a complex query calculating the total cost of treatments for each field.
2. Queries regarding monitoring fertilizer stock.
3. Checking the technical condition of machines.
4. Checking the most active machines.
5. Finding fields with specific topography and soil quality.
6. Comparing the frequency of treatment types.
7. Report on the consumption of a specific pesticide on fields.
8. List of machines that did not work in the current year.
9. Checking unavailable machines.
10. Checking the five oldest machines.

Indexes:

Indexes on foreign keys:

```
CREATE INDEX idx_treat_field ON Treatments(field_id);  
CREATE INDEX idx_treat_machine ON Treatments(machine_id);  
CREATE INDEX idx_pest_unit ON Pesticides(unit_id);
```

Optimization of periodic reports

```
CREATE INDEX idx_treat_date ON Treatments(treatment_date);
Accelerated display of active machines:
CREATE INDEX idx_machine_status ON Machines(status);
```

Views

1. View displaying treatment history with details:

```
CREATE VIEW v_history_details AS
SELECT
  t.treatment_date AS treatment_date,
  f.field_number AS field_no,
  t.treatment_type AS work_type,
  m.machine_name AS machine,
  p.pesticide_name AS agent_name,
  t.amount || ' ' || t.unit_id AS dosage
FROM Treatments t
JOIN Fields f ON t.field_id = f.field_id
JOIN Machines m ON t.machine_id = m.machine_id
LEFT JOIN Pesticides p ON t.pesticide_id = p.pesticide_id;
```

2. Machine overview and fuel consumption analysis:

```
CREATE VIEW v_machine_usage AS
SELECT
  m.machine_name AS machine_name,
  COUNT(t.treatment_id) AS total_treatments,
  SUM(tc.fuel_cost) AS total_fuel_expenses,
  m.status AS technical_status
FROM Machines m
LEFT JOIN Treatments t ON m.machine_id = t.machine_id
LEFT JOIN Treatment_cost tc ON t.treatment_id = tc.treatment_id
GROUP BY m.machine_name, m.status;
```

Users:

Administrator: Everything

Accountant: Sees everything but can change nothing except in the treatments_cost table Machine

Operator: Sees and can edit only machines

Manager: Sees everything and can edit treatments and fields tables

Table normal forms:

- Fields - 3NF
- Machines - 3NF
- Pesticides - 3NF
- Units - 3NF
- Treatments - 3NF
- Treatment_cost - 2NF due to more detailed information